

<activity07 />

In-Class

You will submit only one file to Canvas with name:

`netId_Activity07_functions_objects.js`



The activity has 6 exercises using functions and objects.
You will develop all exercises in the same JavaScript file.



Exercise <1/>

Write a function `maxOfTwo(n1,n2)` that receives two params and it returns the maximum between the params.

INPUT :

```
let n1 = 11;  
let n2 = 10;  
console.log(`The max between ${n1} and ${n2} is :`, maxOfTwo(n1,n2));
```

OUTPUT :

11

Exercise <2/>

Write a function `maxOfArray(array)` that receives one array of numbers as a parameter, and it returns the maximum among all the numbers.

* Do not use `Math.max(...array)`

INPUT:

```
let array = [10,11,1024,125,9,201];  
console.log(maxOfArray(array));
```

OUTPUT :

1024

Exercise <3/>

Given the next object, write a function **showProperties(movie)** to display the list of property names (**keys**) and then the list of property values (**values**):

INPUT:

```
// Object :  
const movie = {  
  title : 'Some movie',  
  releaseYear: 2018,  
  rating: 4.5,  
  director: 'Steven Spielberg'  
};  
  
showProperties(movie);
```

OUTPUT :

List of Keys :	0!
title	0!
releaseYear	0!
rating	0!
director	0!
List of Values :	0!
Some movie	0!
2018	0!
4.5	0!
Steven Spielberg	0!

Hint: **for (let key in myMovie)**
gives you the list of keys.

Exercise <4/>

Create a **circle** object with **key** **radius**, and **value** **2**.

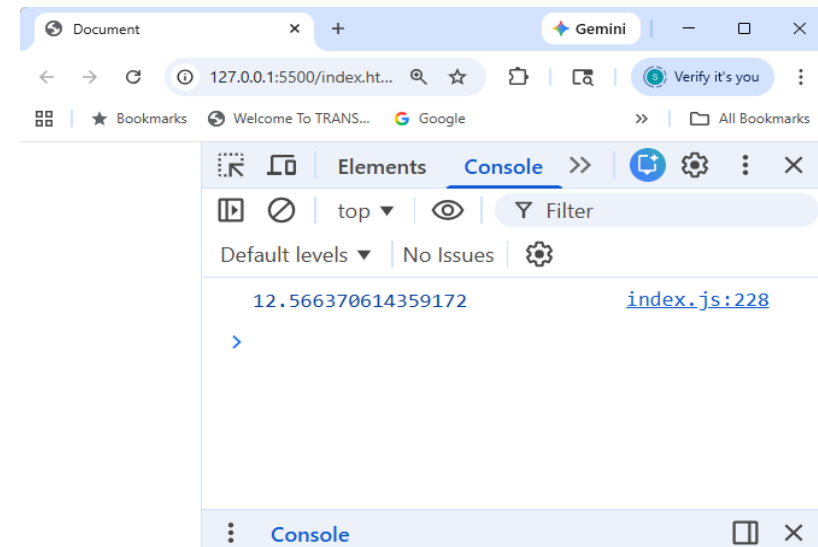
Add a **key** **area** which is a **function()** that computes the area of the circle.

The area of a circle given its radius is : `Math.PI * this.radius * this.radius;`

INPUT:

```
console.log(circle.area());
```

OUTPUT :



Hint: **Math.PI** is equal to 3.141592

Exercise <5/>

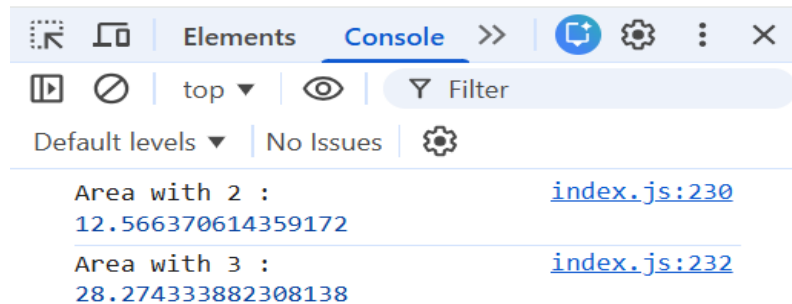
Modify the Exercise 4, create **circle2** object to include **getter** and **setter** for the **key radius**.

INPUT:

```
setter → console.log(`Area with ${circle2.radiusValue} :`, circle2.area());  
circle2.radiusValue = 3;  
getter → console.log(`Area with ${circle2.radiusValue} :`, circle2.area());
```

Observe there is a parenthesis.
Why ?

OUTPUT :



```
Area with 2 : 12.566370614359172  
Area with 3 : 28.274333882308138
```

Observe there is not parenthesis.
Why ?

Hint: use `get radiusValue(){ ... }`
`set radiusValue(value) {...}`

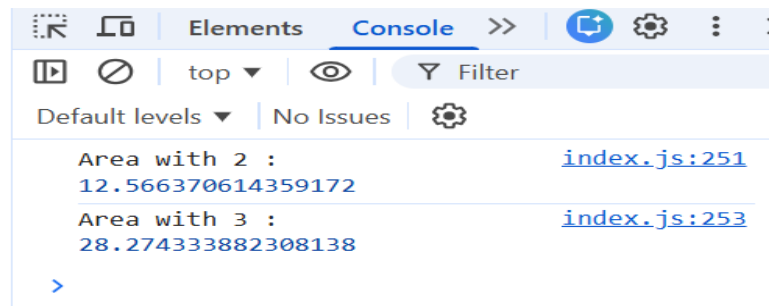
Exercise <6/>

Modify the Exercise 5, create **circle3** object to use functions instead of the **getter** and **setter** to return and set the value of radius respectively.

INPUT:

```
console.log(`Area with ${circle3.getRadiusValue()} :`, circle3.area());
circle3.setRadiusValue(3);
console.log(`Area with ${circle3.getRadiusValue()} :`, circle3.area());
```

OUTPUT :



Observe there is parenthesis.
Why?

Hint: use `getRadiusValue : function(){ ... }`
`setRadiusValue : function(value) {...}`

Optional problems if there is time

Exercise <7/>

Given the object representing a student's grades, write a function “**CalculateAverageGrade(grades)**” that calculates and returns the average grade of the student..

INPUT:

```
const grades = {  
  math: 85,  
  science: 90,  
  history: 75,  
  literature: 88  
};  
  
console.log(calculateAverageGrade(grades));
```

OUTPUT :

Average Grade: 84.5

Exercise <8/> **This is DIFICULT !**

Given the array containing different objects representing students' grades, write a function “**CalculateAverageGrade(array)**” that calculates and returns an object with the average grade for all the students.

INPUT:

```
const students = [
  {
    Fer: {
      math: 85,
      science: 90,
      history: 75,
      literature: 88
    }
  },
  {
    Alex: {
      math: 99,
      science: 97,
      history: 94,
      literature: 90
    }
  },
  {
    Mary: {
      math: 79,
      science: 72,
      history: 81,
      literature: 79
    }
  }
];
```

```
console.log(calculateAverageGrade(grades));
```

OUTPUT :

```
console.log(averageGrades);
```

```
► {Fer: 85, Alex: 95, Mary: 78}
```